

worn like a backpack. It provides astronauts many of the things they need to survive on a spacewalk. Its tanks supply oxygen for the astronauts to breathe. It removes exhaled carbon dioxide. It contains a battery for electrical power.

The PLSS also holds water-cooling equipment, a fan to circulate oxygen and a two-way radio. A caution and warning system in this backpack lets spacewalkers know if something is wrong with the suit. The unit is covered with protective cloth layers.

- ▶ **Layers** – The spacesuit has 14 layers of material to protect the spacewalker. The liquid cooling and ventilation garment makes up the first three layers. On top of this garment is the bladder layer. It creates the proper pressure for the body. It also holds in the oxygen for breathing. The next layer holds the bladder layer to the correct shape around the astronaut's body and is made of the same material as camping tents. The ripstop liner is the tear-resistant layer. The next seven layers are Mylar insulation and make the suit act like a thermos. The layers keep the temperature from changing inside. They also protect the spacewalker from being harmed by small, high-speed objects flying through space. The outer layer is made of a blend of three fabrics. One fabric is water-proof. Another is the material used to make bullet-proof vests. The third fabric is fire-resistant.
- ▶ **Upper Torso** – The top of the spacesuit includes the Hard Upper Torso (HUT) and the arm assembly. The HUT covers the chest and back. It is a vest made out of fibreglass like some cars and swimming pools. The Displays and Control Module and Primary Life Support Subsystem attach to this piece. An important function of this piece is that it serves as the connection for the tubes that drain water and allow oxygen flow.
- ▶ **Displays and Control Module** – This module is the control panel for the EMU, containing switches, controls, gauges and an electronic display. The astronaut can operate the PLSS from this module.
- ▶ **Arms & Gloves** – Spacewalkers do not wear custom-made suits. Different sizes of arm assembly parts are available. Sizing rings can make the parts longer or shorter.

Astronauts must be able to work with and pick up objects while wearing spacesuit gloves. EVA gloves are made to protect astronauts from the space environment. They are also made so spacewalkers can move their fingers as easily as possible. The fingers are the part of the body that gets coldest in space. These gloves have heaters in the fingertips. A piece